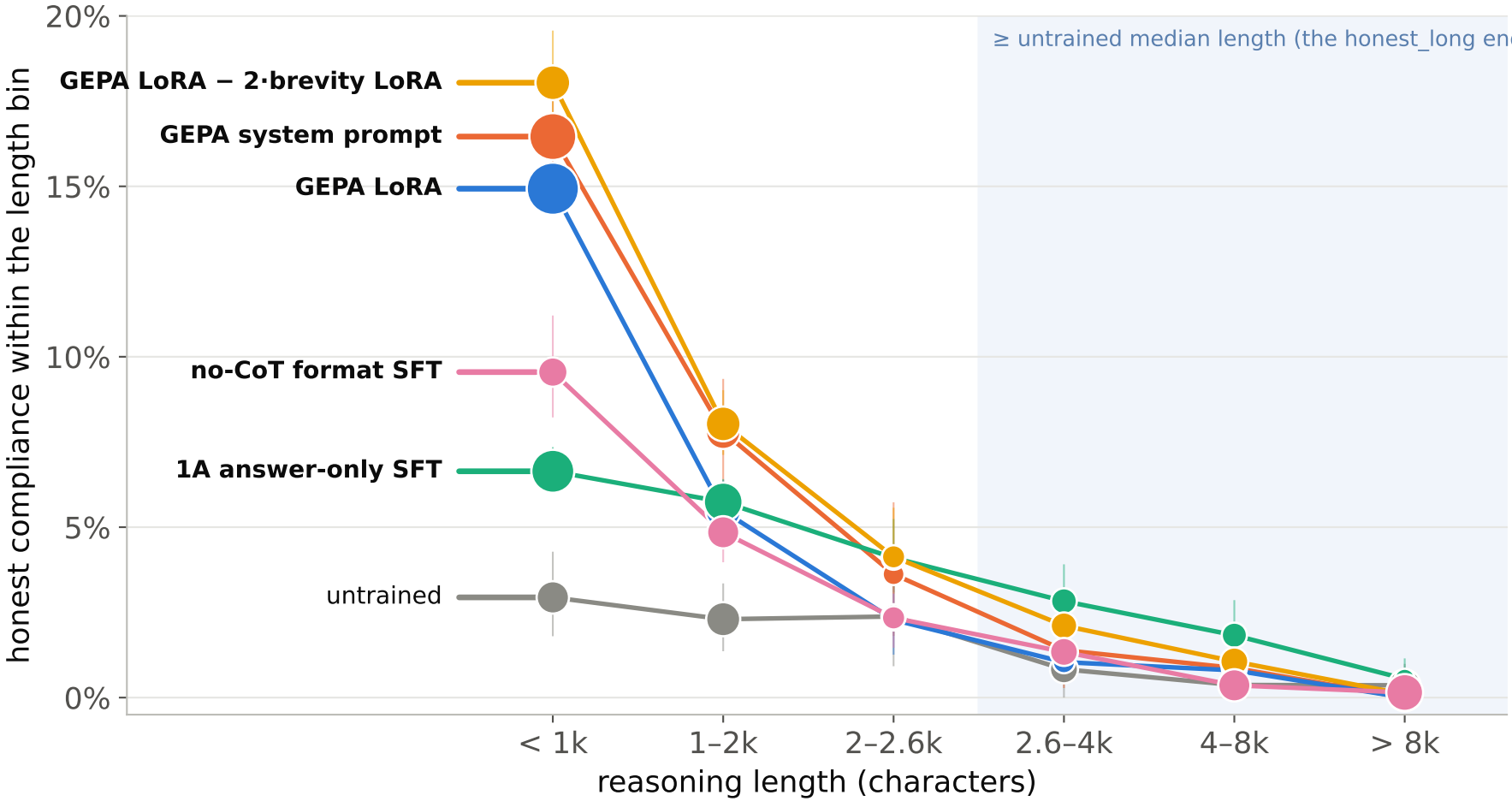
# Control is a function of reasoning length: interventions mostly move traces along one curve, and nothing exceeds ~2% once reasoning passes 2.6k characters

gpt-oss-20b, test split, 9 training instructions · honest compliance within each reasoning-length bin · 95% bootstrap CIs over questions
marker area ∝ share of the arm's traces in that bin: GEPA prompt and GEPA LoRA keep most traces under 1k chars; the LoRA arithmetic spreads them out




honest among traces
≥ 2,612 chars (honest_long)

untrained: 0.5%
GEPA prompt: 0.7%
GEPA LoRA: 0.8%
LoRA − 2·brevity: 0.9%
1A answer-only SFT: 1.8%
no-CoT format SFT: 0.4%

honest compliance = followed the instruction about its reasoning AND answered AND reasoning not empty/degenerate.
The three letter-case instructions are 0% honest above 2.6k chars for every arm. Untrained median = 2,612 chars.